BT230020CS

ET/W-24/ET30 001/03

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

B.Tech ~~Summer~~ / Winter - 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

CSE/IT/AI DS/ME/CE/EE

Branch: Electronics & Telecommunication/ **Sem: III**

Subject: E- Waste Management **Subject Code: ET3O001**

Marks: 100 **Date of Exam: 25/11/2024** **Time: 3 Hrs.**

Instructions:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Use of Programmable Calculator is prohibited.**
4. **Assume suitable data whenever necessary.**
5. **Illustrate your answer whenever necessary with the help of neat sketches.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. **Understand the fundamental concepts and current issues related to e-waste management.**
2. **Apply various principles and practices for e-waste management.**
3. **Analyze the environmental and human health impacts of e-waste and evaluate potential solutions.**
4. **Evaluate the effectiveness of existing national and international regulations for e-waste management.**
5. **Create an informed and responsible plan for e-waste disposal in a specific context.**

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one / two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	What is the composition of e-waste?	L1/CO1	2
Q.2	What is "CRT"?	L1/CO2	2
Q.3	Interpret the term "FICCI".	L2/CO3	2
Q.4	Which state "Silent Valley Movement" happened?	L1/CO3	2
Q.5	Which toxic gas leaked in Bhopal gas tragedy?	L1/CO3	2
Q.6	What is "PCB"?	L2/CO1	2
Q.7	What is "CPCB"?	L1/CO2	2
Q.8	The Basel city is situated in which country?	L1/CO2	2
Q.9	What are the pollutants found in e-waste?	L1/CO1	2
Q.10	Who is the current Union Minister of Environment, Forest and Climate Change (MoEF&CC) in India?	L1/CO2	2

Q.11	Which country join the BASEL convention recently?	L1/CO5	2
Q.12	Write the place where Bishnoi Movement happened.	L1/CO3	2

SECTION B

1. **All Questions are compulsory.**

Q.13	Classify the essential factors in global waste trade economy.	L2/CO2	4
Q.14	Explain in brief Chemical waste and Plastic waste.	L2/CO1	4
Q.15	Identify the impact of global waste trade.	L3/CO2	4
Q.16	Explain in brief Silent valley movement.	L2/CO3	4
Q.17	Identify the effects of pollutant (E- waste) on human health and surrounding environment.	L3/CO1	4
Q.18	Explain in brief basic disposal methods.	L2/CO1	4

SECTION C

1. **Solve any FOUR Questions.**

Q.19	Analyze the guiding principles for fixing up the layout of first level WEEE/E-waste treatment facility.	L4/CO1	7
Q.20	Examine the impact of Italy dumping hazardous chemicals in Nigeria.	L4/CO2	7
Q.21	Justify what you think on India's stand on liberalizing import rules.	L5/CO2	7
Q.22	Explain in brief e-waste economy in the organized sector.	L5/CO2	7
Q.23	Examine the function of EPR (Extended Producers Responsibility) with suitable flow chart.	L4/CO3	7
Q.24	Explain in brief Government assistance for Treatment, Storage and Disposal Facilities (TSDFs).	L5/CO4	7

SECTION D

1. **Solve any TWO Questions**

Q.25	Discuss in detail Waste Electrical and Electronic Equipment (WEEE) Directive in the European Union.	L6/CO5	12
Q.26	Elaborate the E-Waste (Management) Rules 2016.	L6/CO4	12
Q.27	Explain in detail administrative controls & engineering controls with suitable examples.	L5/CO3	12